Bus Reservation and Fleet Scheduling System

DBMS Project — Review 1

Project Details

Team Member

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Department

CSE AI/ML Leopards

Institution

Woxsen University

Course & Deliverable

Course

Database Management Systems

Review Phase

Review 1 — Conceptual & Design Phase

Scope

Relational Database Design for Bus Reservation & Fleet Scheduling

Problem Identification

Current Manual System Issues

Duplicate Seat Allocation

Same seat booked twice due to lack of real-time synchronization across counters.

Schedule Management Failure

Difficulty tracking bus availability and trip schedules leads to operational chaos.

Crew Conflicts

Driver and conductor scheduling overlaps result in unattended trips and disputes.

Cancellation & Refund Errors

Manual cancellations lead to errors, unprocessed refunds, and passenger complaints.

Data Fragmentation

Passenger, booking, and payment data scattered across disconnected systems.

Poor Reporting

No mechanism to generate occupancy, revenue, or fleet utilization reports reliably.

No Data Integrity

Absence of constraints means business rules are unenforced and violations go undetected.

Solution: A structured relational database with integrity constraints (PK, FK, UNIQUE, CHECK) to ensure data consistency and eliminate manual errors end-to-end.

Project Scope

System Components Covered



Route & Stop Management

Define routes, stops, and stop sequences with ordered relationships between nodes.



Bus & Seat Management

Track buses, seat inventory per bus, and real-time seat availability status.



Trip / Schedule Management

Publish trips, manage departure and arrival schedules, and monitor trip status.



Driver & Conductor Management

Assign crew to trips, detect conflicts, and track crew assignments over time.



Booking & Cancellation

Reserve seats with passenger details, manage cancellations, and process refunds.



Fare & Payment

Calculate fares per route and passenger type; record payments and payment methods.



Reporting & History

Generate occupancy, revenue, crew schedule, and fleet utilization reports.



Passenger Management

Maintain passenger profiles, contact information, and booking history records.

Focus: Database design for reservation and fleet scheduling with data integrity and reporting capabilities.

Project Objectives

Core Design & Technical Goals

1 Structured Relational Database

Design following normalization principles up to Third Normal Form (3NF) to eliminate redundancy and anomalies.

2 Eliminate Duplicate Seat Bookings

Enforce UNIQUE constraints and atomic transactions to prevent concurrent double-bookings.

3 Maintain Data Integrity

Apply primary keys, foreign keys, NOT NULL, CHECK, and DEFAULT constraints throughout the schema.

4 Manage Routes, Buses, Seats & Schedules

Establish proper relational associations so all fleet data remains consistent and queryable.

1 Assign Drivers & Conductors

Model crew assignments with conflict detection — no driver or conductor double-assigned to overlapping trips.

2 Track Bookings, Cancellations & Payments

Maintain a complete audit trail for all reservation transactions including refund records.

3 Reporting Capabilities

Support SQL-based reports for occupancy rates, revenue summaries, and fleet utilization analytics.

4 Normalize to 3NF

Remove transitive and partial dependencies across all 14 tables to achieve a clean, anomaly-free schema.

System Users & Roles

Key Actors & Their Responsibilities



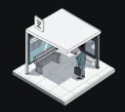
Admin / Bus Operator

Manages buses, routes, and trip schedules. Handles crew assignments, reviews operational reports, and maintains system configuration.



Booking Staff / Counter Operator

Searches available trips, books seats on behalf of passengers, processes cancellations, and handles payment collection at counters.



Passenger

Searches for trips by route and date, books seats, views personal booking history, and initiates cancellation requests.



Driver

Views assigned trips, departure times, and route information. Read-only access to relevant scheduling data.



Conductor

Views crew assignments, trip details, and passenger manifest information for the assigned trip.

Each role has specific permissions and access to relevant data through the database system.

Functional Requirements

Reservation Module

- Search available trips by route, date, and time
- View available seats for a selected trip
- Book a seat with full passenger details
- Cancel booking and process refunds automatically

Fleet Management

- Add and manage buses with capacity and type
- Manage seat inventory per bus
- Track bus availability and maintenance status

Scheduling & Crew

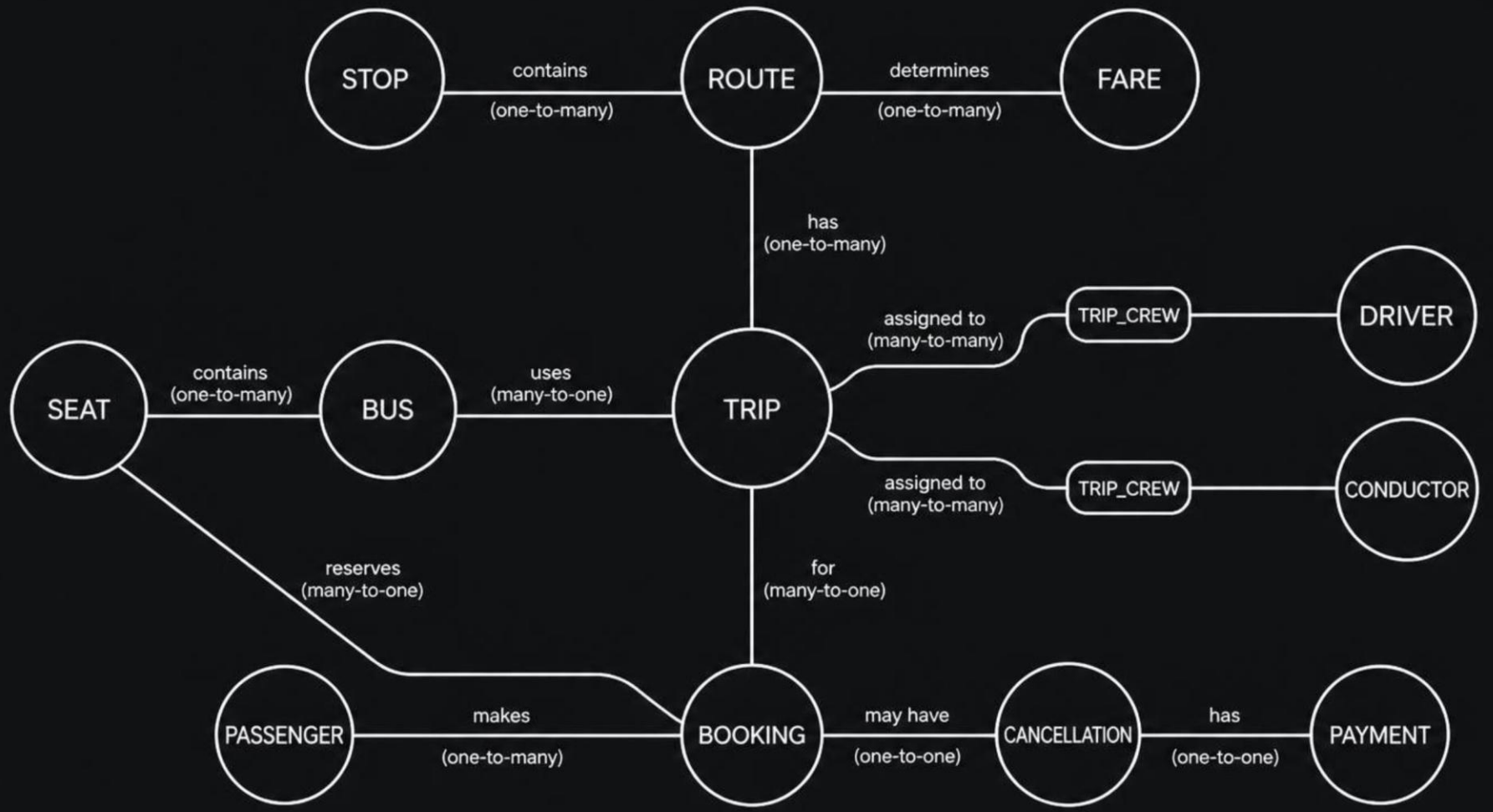
- Publish trips and manage schedules
- Assign drivers and conductors to trips
- Manage trip status: **scheduled → in-progress → completed / cancelled**

Finance & Reporting

- Calculate fares based on route and passenger type
- Record payments and payment methods
- Generate occupancy, revenue, and crew schedule reports

Conceptual Design — ER Diagram

Principal Entities and Relationships (Crow's Foot Notation)



Entities: ROUTE, STOP, BUS, SEAT, TRIP, DRIVER, CONDUCTOR, PASSENGER, BOOKING, CANCELLATION, FARE, PAYMENT

Key Entities & Attributes

Entity Definitions with Primary Keys, Foreign Keys & Constraints

Entity	Key Attributes
ROUTE	route_id (PK), source, destination
STOP	stop_id (PK), stop_name, location
BUS	bus_id (PK), registration_no (UNIQUE), bus_type, capacity
SEAT	seat_id (PK), bus_id (FK), seat_number
TRIP	trip_id (PK), route_id (FK), bus_id (FK), departure_time, arrival_time, status
DRIVER	driver_id (PK), name, license_no (UNIQUE), phone
CONDUCTOR	conductor_id (PK), name, phone
PASSENGER	passenger_id (PK), name, phone, email
BOOKING	booking_id (PK), trip_id (FK), passenger_id (FK), seat_id (FK), booking_date, status
CANCELLATION	cancellation_id (PK), booking_id (FK), cancellation_time, refund_amount
FARE	fare_id (PK), route_id (FK), amount
PAYMENT	payment_id (PK), booking_id (FK), amount, payment_method, payment_status

Initial Relational Schema

14 Tables — PK (underlined), FK (in brackets), Key Constraints Noted

ROUTE & STOPS

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ROUTE(
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Review 1 Summary & Next Steps

✓ Completed in Review 1

- Problem identification and motivation clearly defined
- Project scope and eight system components established
- System users and functional requirements identified
- Principal entities and relationships documented
- Conceptual ER model designed (Crow's Foot notation)
- Initial relational schema proposed — 14 tables

Next Steps — Review 2 & Beyond

01

Normalize to 3NF

Remove all partial and transitive dependencies; finalize the relational schema.

02

Add Full Constraint Set

Implement PK, FK, UNIQUE, NOT NULL, CHECK, and DEFAULT constraints in DDL.

03

Implement Database

Create schema in MySQL/PostgreSQL; insert realistic sample data for testing.

04

Develop SQL Queries

Write joins, aggregates, nested queries, stored procedures, triggers, and views.

05

Front-End & Testing

Build application interface (Java/Python); test booking, cancellation, and transaction scenarios end-to-end.